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Logistic Regression

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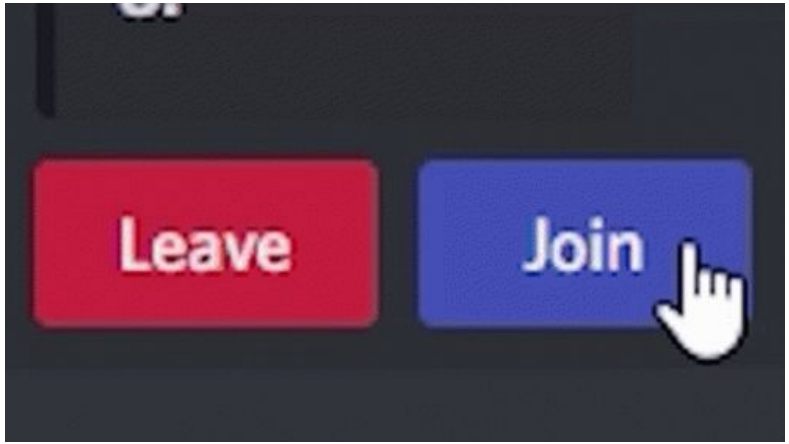
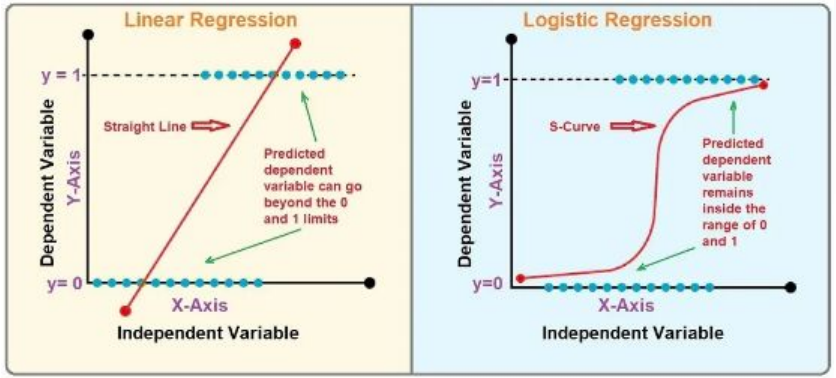


Recap: Linear Regression

The Basics

- **Brief Reminder:** Linear Regression predicts continuous numbers (e.g., Salary, Temperature).
- **The Problem:** A straight line fails to categorize data into discrete "Yes/No" groups effectively.
- **Key Question:** "How do we fit a line when the answer is just 0 or 1?"



The Classification Challenge

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Definition: Predicting a category rather than a number.

Purchase?

Will the customer buy?
(Yes / No)

Spam Filter

Is this email Spam?
(Spam / Not Spam)

Academics

Will the student pass?
(Pass / Fail)

Why Linear Regression Fails Here

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1. Unbounded Predictions

Predictions can go above 1 or below 0 (e.g., -0.2 probability), which is mathematically impossible for probabilities.

2. Bad Fit

The "Straight Line" doesn't fit the "Step" nature of binary decisions, leading to poor accuracy and high error rates near the decision boundaries.

Enter Logistic Regression

The Solution

We need a line that curves to stay strictly between 0 and 1.

Definition: A regression analysis used for predicting the probability of a binary outcome.

It answers: "What is the % chance this is a 'Yes'?"

The Magic "S" Curve (Sigmoid)

The Concept

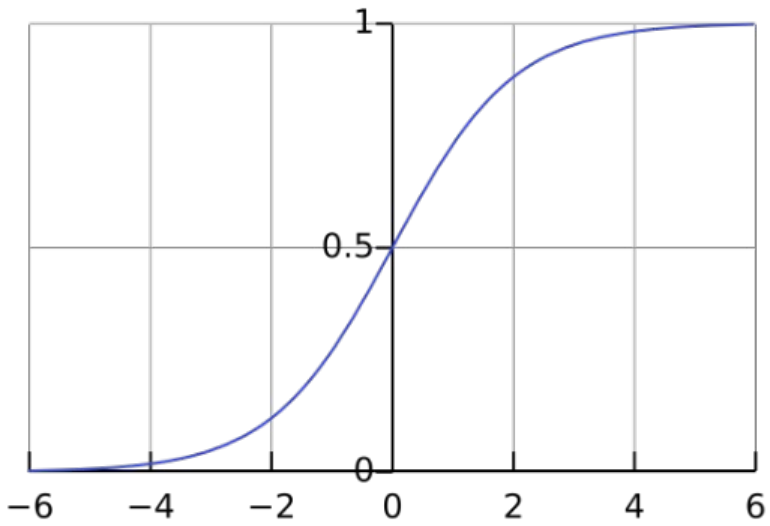
To fix the "infinity" problem, we use the Sigmoid Function. It takes any number (big or small) and squashes it into a value strictly between 0 and 1.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Visualizing

The S-shape curve crosses the center at 0.5.

- High Positive Value → Probability near 1
- High Negative Value → Probability near 0



The "Log-Odds" Concept

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The Output

The model gives a decimal probability (e.g., 0.75). How do we decide Yes or No?

Usually set at a **Decision Boundary** of 0.5.

- Score > 0.5 = Class 1 (Yes)
- Score < 0.5 = Class 0 (No)

Beyond Values

Unlike Linear Regression, we don't predict the value directly. We predict the **Log of the Odds** of success.

"If the odds of winning are 3:1, Logistic Regression helps us calculate the probability (75%)."

The Equation

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Linear Regression

The standard line equation:

$$y = mx + b$$

Logistic Regression

We wrap the linear equation inside the S-curve function!

$$y = \sigma (mx + b)$$

Practical Question 1: The Boundary

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Scenario

Context: Spam Filter.

Data: Model score = 0.3.

Threshold: 0.5.

Question: Inbox or Spam? (1 = Spam)

Answer: Inbox

Since 0.3 < 0.5, it is classified as Class 0 (Not Spam).

Inbox

Practical Question 2: Variable Impact

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Scenario

Context: Heart Disease Prediction.

Variable: "Hours of Exercise" (x).

Coefficient: -0.5 (Negative).

Question: Does more exercise increase or decrease probability?

Answer: Decrease

A negative weight means as **x** goes up, the probability goes down.

Practical Question 3: Outliers

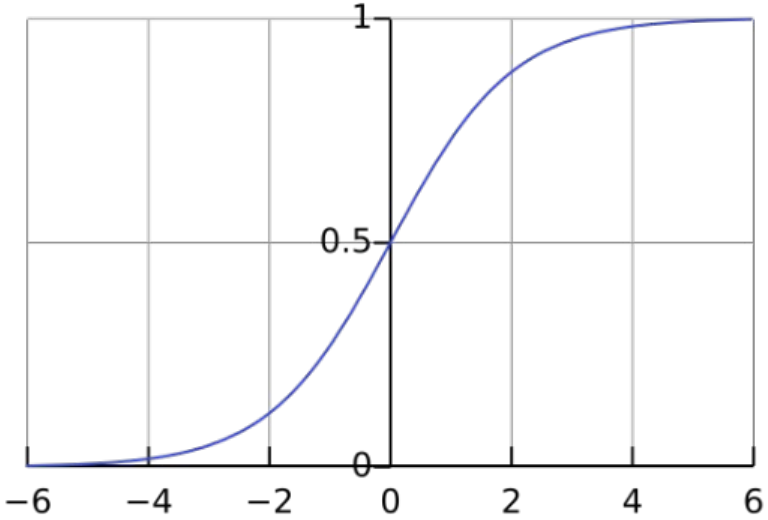
Scenario

Context: Salary Dataset.

Data: One person makes $1 Billion/year (Extreme Outlier).

Why Logistic is Better?

The Sigmoid function "squashes" the outlier. Whether input is 1 million or 1 billion, the curve caps probability at 1.0, preventing the model from breaking.



Practical Question 4: Probabilities

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Scenario: Weather App. Output: P(Rain) = 0.5.

0.5

Maximum Uncertainty

The model is completely unsure. It acts like a 50/50 coin toss.

Practical Question 5: Thresholding

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Scenario

Context: Cancer Detection AI.

Question: Should we keep the threshold strictly at 0.5?

Discussion: No

If missing a diagnosis is dangerous, we lower the threshold (e.g., to 0.3) to catch more potential cases, accepting more false alarms (False Positives) to avoid missing actual cases (False Negatives).

Case Study: University Admission

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Predicting Admission

Problem: Predict if student gets into Harvard (Yes/No).

Features: GPA, SAT Score.

Logic:

- High GPA → Increases Probability
- Low SAT → Decreases Probability

Practical Question 6: Interpretation

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Scenario

In the Admission model, the "Extracurriculars" feature has a coefficient of **0**.

Interpretation

It has **no effect** on the prediction.

The model has learned that this specific feature does not help determine the outcome.

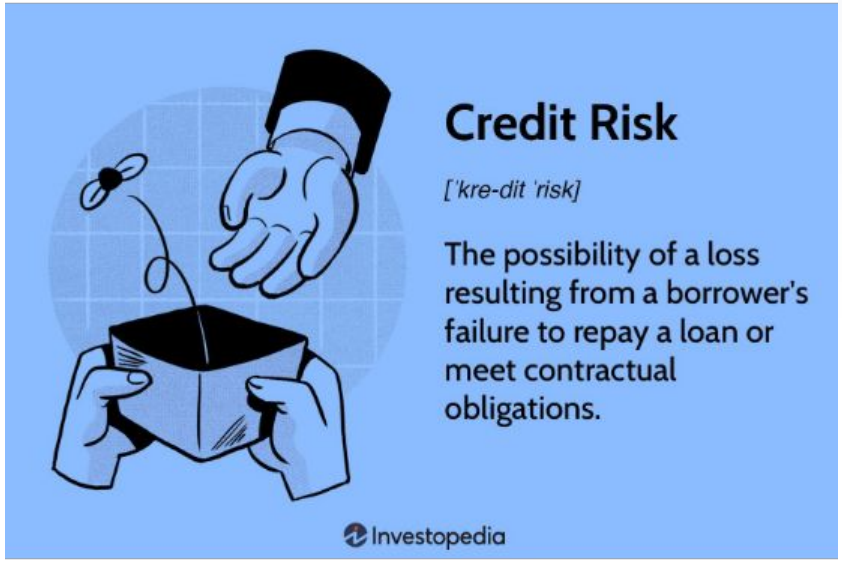
Case Study: Credit Card Default

Risk Model

Will a customer pay back their loan? (S=Salary, D=Debt)

We plug numbers into the equation. Lower (more negative) output means higher risk.

- **Person A:** High Salary, High Debt.
- **Person B:** Low Salary, No Debt.



Practical Question 7: Model Learning

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Scenario

We train the model and it predicts "Class 1" for everything.

Diagnosis: Imbalanced Data

If 99% of data is "Class 1", the model becomes "lazy" and guesses "1" every time to achieve high accuracy without actually learning patterns.

Practical Question 8: Non-Linearity

Scenario

You plot your data, and "Yes" dots are in the middle, surrounded by "No" dots in a circle.

Can Logistic Solve This?

No. Logistic Regression draws a straight line boundary. It cannot draw a circle unless we add complex polynomial features.

Assumptions (What you need)

- **Binary Outcome:** The target variable must strictly have only two categories.
- **Independence:** Data points shouldn't influence each other (e.g., distinct customers).
- **No Multicollinearity:** Predictor variables shouldn't be copies of each other (e.g., "Weight in Kg" and "Weight in Lbs").

Pros & Cons

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Pros

- Easy to interpret (we know which features matter).
- Gives probabilities, not just rigid labels.
- Efficient to train on large datasets.

Cons

- Struggles with complex, non-linear patterns.
- Sensitive to outliers (though better than linear).
- Requires clean data (no multicollinearity).

Comparison Table

Feature	Linear Regression	Logistic Regression
Output Type	Continuous Number	Probability (0 to 1)
Visual Shape	Straight Line	S-Curve (Sigmoid)
Range	$(-\infty, +\infty)$	$(0, 1)$

Final Review Quiz

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What is the range of the Sigmoid function?

0 to 1

What is the default threshold?

0.5

Can it predict 3 classes?

Not the basic version

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Questions?

Thank you for your attention.

Next Step: Neural Networks

Basically many Logistic Regressions stacked together!